Lesson 14 -> Load Balancers

Load Balancers

1️⃣ What is a Load Balancer?

A **Load Balancer** is a system that **distributes incoming network traffic across multiple servers** to ensure that no single server becomes overloaded.

Instead of sending all user requests to one server, a load balancer spreads the traffic across several servers.

Basic Architecture

Users → Load Balancer → Multiple Servers

```
    Users
      |
      v
Load Balancer
 /     |     \
v      v      v
```

Server1 Server2 Server3

The load balancer acts as a **traffic manager** between clients and servers.

2️⃣ Advantages of Load Balancers

High Availability

If one server fails, traffic is redirected to other healthy servers.

Server1 ❌
Server2 ✅
Server3 ✅

Users still get responses.

Scalability

When traffic increases, we can **add more servers** behind the load balancer.

Server1
Server2
Server3
Server4
Server5

The system can scale horizontally.

Improved Performance

Traffic is distributed across servers, reducing the load on each server and improving response time.

Fault Tolerance

If one server crashes, the load balancer automatically routes requests to healthy servers.

Better Resource Utilization

All servers participate in serving traffic instead of a single server handling everything.

3️⃣ Types of Load Balancers

Load balancing strategies are broadly classified into:

```
Load Balancing Algorithms
|
|--- Static
|
|--- Dynamic
```

Type	Description
Static Load Balancing	Decision based on predefined rules
Dynamic Load Balancing	Decision based on real-time server state

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4️⃣ Static Load Balancers

In **static load balancing**, the algorithm distributes traffic **without considering the current server load**.

It assumes that all servers have similar capabilities.

Characteristics

- No real-time monitoring
- Simple implementation
- Works well when servers are identical

Static Load Balancing Algorithms

1️⃣ Round Robin

Requests are distributed sequentially across servers.

Example:

Request1 → Server1
Request2 → Server2
Request3 → Server3
Request4 → Server1
Request5 → Server2

Advantages

- Very simple
- Equal distribution of traffic

Disadvantages

- Does not consider server capacity
- Slow servers may become overloaded

2️⃣ Weighted Round Robin

Each server is assigned a **weight based on its capacity**.

Servers with higher weights receive more traffic.

Example:

Server1 (Weight 3)
Server2 (Weight 2)
Server3 (Weight 1)

Traffic distribution:

Server1
Server1
Server1
Server2
Server2
Server3

Advantages

- Supports servers with different capacities
- Better distribution than basic round robin

Disadvantages

- Still does not consider real-time load

3️⃣ IP Hash

The load balancer chooses a server based on the **client's IP address**.

server_index = hash(client_ip) % number_of_servers

Example:

User IP → Same Server

This ensures **session persistence**.

Advantages

- Same user always goes to the same server
- Useful for session-based applications

Disadvantages

- Uneven traffic distribution if some users generate heavy load

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5️⃣ Dynamic Load Balancers

Dynamic load balancing makes decisions **based on real-time server metrics**.

It considers factors like:

- active connections
- CPU usage
- memory usage
- response time

This leads to more intelligent traffic distribution.

Dynamic Load Balancing Algorithms

1️⃣ Least Connections Method

The request is sent to the server with the **fewest active connections**.

Example:

Server1 → 10 connections
Server2 → 4 connections
Server3 → 7 connections

Next request goes to:

Server2

Advantages

- Better distribution for long-lived connections
- Adapts to real-time traffic

Disadvantages

- Requires tracking server connection counts

2️⃣ Least Response Time

Requests are routed to the server with the **lowest response time**.

The load balancer monitors how fast each server responds.

Example:

Server1 → 200 ms
Server2 → 120 ms
Server3 → 150 ms

Next request goes to:

Server2

Advantages

- Optimizes user latency
- Avoids slow servers

Disadvantages

- Requires continuous monitoring

3️⃣ Resource-Based Load Balancing

Traffic is routed based on **server resource utilization**.

Metrics may include:

- CPU usage
- memory usage
- network utilization

Example:

Server1 → CPU 80%
Server2 → CPU 30%
Server3 → CPU 50%

Next request goes to:

Server2

Advantages

- Most intelligent distribution
- Prevents server overload

Disadvantages

- Complex implementation
- Requires monitoring agents on servers

6️⃣ Static vs Dynamic Load Balancing

Feature	Static	Dynamic
Decision Basis	Predefined rules	Real-time server metrics
Complexity	Simple	More complex
Performance	Good for small systems	Better for large systems
Monitoring	Not required	Required
Adaptability	Low	High

7️⃣ Real-World Load Balancers

Common production load balancers include:

Load Balancer	Type
Nginx	Reverse proxy + load balancer
HAProxy	High-performance load balancer
AWS Elastic Load Balancer	Managed cloud load balancer
Google Cloud Load Balancer	Cloud-native load balancing
Envoy	Modern service mesh load balancer